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MzansiRide SafeAI

A Safety-First, AI-Powered Ride-Sharing Solution for South Africa

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Hackathon Theme: AI for Real-World Impact
Focus: Build. Innovate. Solve.
Cloud Platform: Amazon Web Services (AWS)
Prototype Type: Responsive web application

1. Executive Summary

MzansiRide SafeAI is a safety-first, AI-powered transport prototype designed to address passenger safety concerns and transport affordability in South Africa.

Concerns about gender-based violence (GBV), kidnapping, travelling with unfamiliar drivers, and unexpected journey events can make using transport services stressful for passengers.

MzansiRide SafeAI aims to provide passengers with additional visibility, check-in options, trusted-contact sharing, and AI-assisted safety guidance throughout a journey.

Unlike a conventional ride-booking application that primarily focuses on connecting passengers with drivers, MzansiRide SafeAI places passenger awareness and support at the centre of the journey. The prototype will demonstrate how a passenger can request a ride, activate Safe Journey Mode, receive a prompt when an unusual event occurs, and access guidance through an AI-powered assistant.

The platform will also demonstrate a basic shared-ride fare comparison to explore how passengers travelling in similar directions could potentially reduce transport costs.

The prototype will use simulated trips and predefined safety events to demonstrate the core concept without requiring live GPS tracking, real emergency-service integrations, or a complete commercial transport system.

2. Problem Statement

Transport is essential for people travelling to work, educational institutions, healthcare facilities, and other destinations across South Africa. However, personal safety remains a concern for passengers, particularly when travelling alone or with unfamiliar drivers.

Women and other vulnerable passengers may worry about unexpected route changes, prolonged stops, the identity of a driver, or whether someone they trust knows the details of their journey. While transport applications may provide certain safety features, there is an opportunity to demonstrate a more integrated experience that brings journey monitoring, passenger check-ins, and safety guidance together.

Transport costs are also a challenge for many people, particularly daily commuters and students who depend on affordable travel options.

The challenge is to explore how AI and cloud technology can help passengers feel more informed and supported during their journeys while demonstrating practical opportunities for shared transport.

3. Proposed Solution

MzansiRide SafeAI will be a responsive web application that demonstrates a safer approach to ride-sharing.

Passengers will be able to enter pickup and destination details, request a simulated ride, view driver and vehicle information, and activate Safe Journey Mode.

During the simulated journey, the system will detect predefined events, such as an unexpected, prolonged stop or a route deviation. When an event is detected, the application will ask the passenger to confirm whether everything is okay and display available assistance options.

An AI-powered safety assistant, built using Amazon Bedrock, will explain the alert and provide relevant guidance based on approved information. The system will also demonstrate how an alert can be recorded and how a trusted-contact notification workflow could operate.

A basic fare comparison will illustrate how shared rides may help reduce travel costs.

The prototype will focus on demonstrating one complete passenger journey, from booking to safety-event handling. It will not attempt to replicate every feature of an established ride-hailing service.

4. Target Users

The proposed solution is intended for:

- **Women and vulnerable passengers:** People who want additional safety support during transport.
- **Students:** People who need accessible and potentially more affordable transport.
- **Daily commuters:** People who regularly travel to work or educational institutions.
- **Budget-conscious passengers:** People interested in shared rides and potential savings.

- **Drivers:** Transport providers whose displayed details and journey status are available to passengers.
- **Transport operators:** Organisations that may benefit from clearer visibility of trip and alert statuses.

For the hackathon, the team will use simulated passenger and driver profiles rather than collecting real users' sensitive information.

5. Core Features

5.1 Passenger Booking Interface

Passengers can enter a pickup point and destination and request a simulated ride.
The interface will display the trip status and estimated fare. Actual transport bookings and payments will not be processed.

5.2 Driver and Vehicle Details

The application will display sample driver information, vehicle details, and a clearly labelled demonstration status.
The prototype will not claim to verify a driver's identity or vehicle unless a real verification process has been implemented.

5.3 Safe Journey Mod

Passengers can activate Safe Journey Mode to view their current simulated trip status and access safety-related options
The journey screen will provide a clear view of the trip, check-in status, and available assistance actions.

5.4 AI-Assisted Safety Monitoring

The prototype will use predefined rules to detect two simulated events:

- An unexpected, prolonged stop.
- A significant deviation from the planned route.

When an event is detected, the system will display an alert and ask the passenger to check in.
The system will not automatically assume that the driver is acting maliciously. Events will be treated as prompts for passenger confirmation, since unexpected stops and route changes can have legitimate explanations.

5.5 AI Safety Assistant

Amazon Bedrock will power an assistant that explains safety alerts and provides guidance based on approved information.
For example, when a simulated trip stops unexpectedly, the assistant can explain why the application has requested a check-in and outline the passenger's available options.
The AI assistant will support decision-making but will not claim to know whether a passenger is safe or predict criminal behaviour.

5.6 Trusted-Contact Notification Demonstration

The application will demonstrate a workflow for selecting a trusted contact and preparing a notification about a journey event.

For the 24-hour prototype, notifications can be simulated and their status displayed on the dashboard. Real notifications can be added if time permits and the required integrations are available.

5.7 Safety Alert Dashboard

A simple dashboard will display simulated trips, detected events, passenger check-in responses, and alert statuses.

This will allow the team to demonstrate how the backend records and processes safety events.

5.8 Shared-Ride Fare Comparison

The application will compare an illustrative individual fare with a shared-ride fare.

The comparison will use simple predefined pricing rules and sample data, demonstrating the potential for shared rides to improve affordability without requiring a complete ride-matching engine.

6. Where AI Adds Value

AI will be used to provide a meaningful capability rather than simply being added as a chatbot.

6.1 Explaining Safety Events

The application will use predefined backend rules to identify a simulated safety event. Amazon Bedrock will then help explain the event in clear, accessible language and describe the available options.

This approach separates event detection from AI-generated explanations, making the prototype more predictable and easier to test.

6.2 Providing Contextual Safety Guidance

The AI assistant will answer questions about the application's safety features and explain the available check-in and assistance workflows using approved guidance.

It will not replace emergency services or make unsupported claims about the cause of an event.

6.3 Supporting Better Passenger Decisions

By presenting the event, explaining what it means, and clearly displaying the available actions, the system aims to help passengers understand their options.

The prototype will demonstrate this workflow using simulated events and responses.

7. Proposed AWS Architecture

The solution will use a small number of AWS services to keep the implementation achievable within 24 hours.

7.1 Frontend Technologies

Technology	Purpose
React.js	Builds the booking, journey, and dashboard screens
TypeScript	Improves code reliability and maintainability
HTML5 and CSS	Provides the web application's structure and styling
Tailwind CSS	Supports responsive layouts and consistent styling
AWS Amplify Hosting	Potential option for hosting the frontend on AWS

Git and GitHub	Supports source-code management and team collaboration

The team will build a responsive web application that works on desktop and mobile-sized screens. Separate native mobile applications will not be developed during the hackathon.

7.2 AWS Services

AWS service	Role in the prototype
Amazon API Gateway	Provides API endpoints for frontend requests
AWS Lambda	Processes simulated trips, safety rules, and alert workflows
Amazon Bedrock	Generates contextual explanations and safety guidance
Amazon DynamoDB	Stores trip and alert records if persistence is included
Amazon CloudWatch	Helps monitor backend logs and errors
Amazon Cognito	Optional authentication if time permits
Amazon SNS	Optional real notification integration if time permits

The team should confirm service permissions, model access, regional availability, and any applicable costs before development begins.

7.3 How the Architecture Works

1. The passenger opens the React web application.
2. The passenger requests a simulated ride.
3. The frontend sends the request to Amazon API Gateway.
4. AWS Lambda processes the request and applies the predefined safety rules.
5. When an unusual event is simulated, Lambda records the event and prepares a response.
6. If an AI explanation is required, Lambda invokes Amazon Bedrock.
7. Trip and alert records can be stored in Amazon DynamoDB.
8. The application displays the alert, passenger check-in prompt, and AI-generated guidance.
9. Amazon CloudWatch provides logs to help the team troubleshoot and validate the demonstration.

Privileged AWS credentials will not be placed in frontend code. Backend operations will be handled through appropriately secured AWS services.

8. Prototype Scope

The project will focus on delivering a working minimum viable prototype within 24 hours.

8.1 Features Included in the Initial Build

The team will aim to deliver:

1. A responsive passenger booking interface.
2. Simulated driver and vehicle details.
3. A journey screen with Safe Journey Mode.

4. Two predefined safety-event scenarios.
5. A passenger check-in prompt.
6. An Amazon Bedrock-powered safety assistant.
7. An alert dashboard.
8. A simulated trusted-contact notification workflow.
9. A basic individual-versus-shared fare comparison.
10. An AWS-backed workflow demonstrating how the components interact.

If the team encounters integration challenges, the essential deliverable will remain the end-to-end safety workflow, with trip data initially simulated or stored locally during early development.

8.2 Features Excluded from the Initial Build

The following will be considered future enhancements:

- Continuous live GPS tracking
- Real-time route deviation detection using a mapping integration.
- Full driver identity and vehicle verification.
- Live emergency-service integration.
- Real payments and driver payouts.
- Production-grade ride matching.
- Native Android and iOS applications.
- Advanced predictive analytics.

These features require additional integration, testing, operational processes, and safety considerations beyond a typical 24-hour hackathon.

8.3 Demonstration Scenario

The team will demonstrate the following sequence:

Step 1: A passenger opens the application and requests a simulated ride.
Step 2: The application displays the sample driver and vehicle information.
Step 3: The passenger activates Safe Journey Mode.
Step 4: The team triggers a simulated prolonged stop or route deviation.
Step 5: The system displays an alert and asks the passenger to confirm whether everything is okay.
Step 6: The AI assistant explains the event and the passenger's available options.
Step 7: The application records the passenger's response and displays the alert status.
Step 8: The team demonstrates the trusted-contact notification workflow and shared-ride fare comparison.

This sequence will provide a clear demonstration of the problem, the proposed solution, the role of AI, and the AWS implementation.

9. Expected Impact

MzansiRide SafeAI aims to demonstrate how AI and cloud services can support more informed passenger decisions and clearer safety workflows.

The prototype will be evaluated using practical indicators, including:

- Whether predefined simulated events are detected correctly.
- The time taken to display a safety prompt after an event is triggered.
- Whether alert and notification statuses are represented accurately.
- The clarity and relevance of AI-generated explanations.
- Whether the passenger can complete the booking and check-in workflow.
- Whether shared-ride fare calculations produce the expected results.

The project aims to raise awareness of safety-focused transport design and demonstrate a potential approach to passenger support. It will not claim to prevent crime or guarantee passenger safety.

10. Privacy, Security, and Responsible AI

The project will use simulated data wherever possible and avoid collecting real passenger location information during the hackathon.

The team will:

- Use sample driver, passenger, and trip records.
- Limit the information stored in the prototype.
- Ensure that privileged AWS credentials are not exposed in frontend code.
- Apply appropriate access controls to backend services.
- Clearly label simulated alerts and notifications.
- Avoid automatically assigning blame to drivers based on unusual journey events.
- Use approved safety information when generating AI responses.
- Explain that AI guidance does not replace emergency services.
- Consider POPIA when designing future data collection and storage.

A production version would require additional security reviews, privacy assessments, driver verification, operational partnerships, transport compliance, and tested emergency-response procedures.